

Mechanics Of Materials  
Basic Mech. Engg.

Mechanics Of Materials  
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Lloads  
●●●●●●

Stress  
●●●●●

Strain  
●●●●●



# Loads

## INTERNAL LOAD IN TRUSSES

**A truss is a structure composed of slender members joined together at their end points**





[1]

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, Fig 6-2, pp. 262

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# Loads

## INTERNAL LOAD IN TRUSSES



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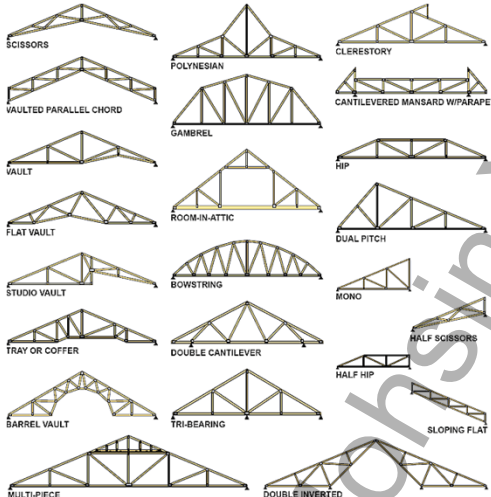
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# Loads

## INTERNAL LOAD IN TRUSSES



[1]

[1] – Website: [http://www.classictruss.com/roof\\_truss\\_facts.asp](http://www.classictruss.com/roof_truss_facts.asp) Visited On: 10-05-2015

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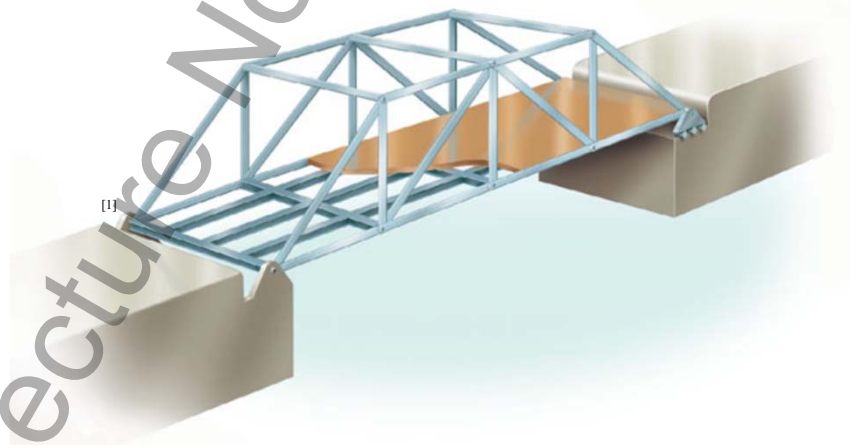
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## INTERNAL LOAD IN TRUSSES



[1]

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, Fig 6-2, pp. 264

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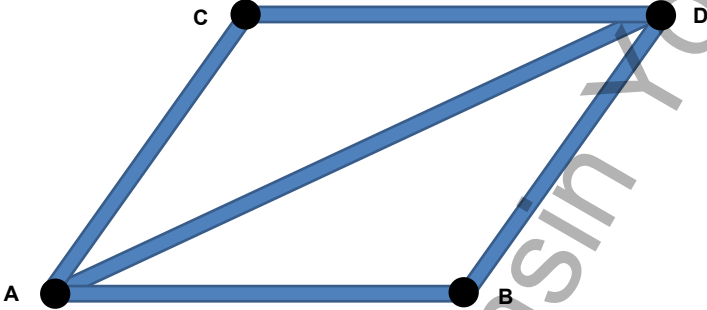
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## INTERNAL LOAD IN TRUSSES



Simplest form that is rigid or stable is a triangle

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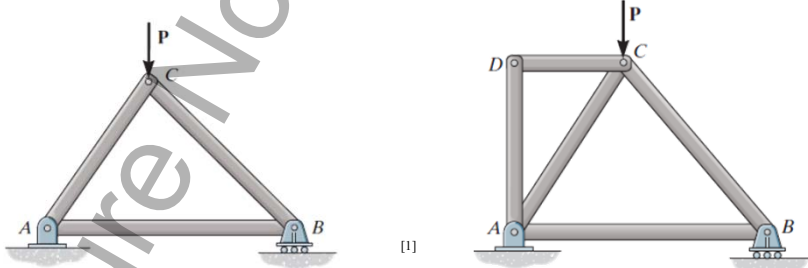
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# Loads

## INTERNAL LOAD IN TRUSSES



A Simple Truss is constructed by starting with a basic triangular element ABC and connecting two members AD and DC to for an additional element.

In order to analyze or design a truss, it is necessary to determine the force in each of its members

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, Fig 6-5, 6-6, pp. 265

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Logo

# Loads

## INTERNAL LOAD IN TRUSSES - METHOD OF SECTIONS

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, Fig 6-15(a), pp. 280

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## INTERNAL LOAD IN TRUSSES - METHOD OF SECTIONS

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, Fig 6-15(b & c), pp. 281

**Cut Section Should not intersect more than 3 members!**  
**But Why?**

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